

Leadership competencies for digital transformation: An exploratory content analysis of job advertisements

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Abstract

Harnessing the opportunities of emerging information technologies is one of the great challenges companies are facing today. To successfully master digital transformation, organizations need leaders who can grasp the opportunities of digitalization for their business and transform them into new business models. Aiming at providing empirical evidence regarding competencies sought by practitioners for managing digital transformation, we analyze 239 job advertisements targeting digital transformation experts and examine the skills and traits explicitly called for. Our results reveal that technical skills and in-depth expertise in information technologies play only a secondary role in job requirement profiles. Like in earlier strategic change processes, digital transformation experts should primarily possess collaboration, strategic thinking, leadership, customer orientation, and communication skills. Moreover, in contrast to purely conceptual studies limited to skills and abilities, our analyses emphasize the importance of specific personality traits, such as proactivity and creativity, since these are often explicitly mentioned in corresponding job advertisements.

Keywords

Change management, digital transformation, job advertisements, leadership, personality traits, skills

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Introduction

Due to rapid developments in digital technology, we are facing a new era characterized by high uncertainty and disruption. In the working environment, these developments based on advanced information and communication technology (ICT) recently associated with keywords such as internet of things, industry 4.0, augmented reality, machine learning, artificial intelligence, and big data spur extensive transformations of organizations (Porter and Heppelmann, 2015; Tapscott, 2015; Weill and Woerner, 2015). As the pace of digital transformation is expected to even accelerate in the future, it is challenging the survival of many firms and industries (Bawany, 2020; Fitzgerald et al., 2013; Hess et al., 2016).

Digital transformation (DT), defined as “*a process that aims to improve an entity by triggering significant changes to its properties through combinations of information, computing, communication, and connectivity technologies*” (Vial, 2019: 3), is characterized by velocity and a holistic nature (Hartl and Hess, 2017). DT does not only affect product and service offerings of organizations, but their organizational processes, systems, and operations, too (e.g., Bohnsack et al., 2018; Strohmeier, 2020; Vial, 2019). Consequently, it requires an organization-wide change program, which embraces all impact factors from new strategic initiatives to detailed human resource management measures such as personnel selection, incentivization, or specialized training, to be successful.

By enabling managers and leaders¹ to rethink and to reshape for example customer value propositions and to renew organizational structures and processes resulting in the creation of new business models (Berman, 2012; Hess et al., 2016), DT can radically improve a company's performance (Bekkhuss, 2016; Vale-Feigl, 2021). This poses major challenges for managers. Not only do they have to monitor technological developments and decide which of them represent a threat or an opportunity to their business, but at the same time they must evaluate how they might alter customer needs, product and service specifications, and consequently organizational structures and processes. Such a comprehensive and far-reaching change process furthermore involves the management of paradoxes such as the active involvement of employees who resist organizational and individual changes (Kane et al., 2019; Miller and Dess, 1996).

Against this backdrop, previous studies have highlighted the importance of developing digital competencies for future business success (Hoberg et al., 2015; Sia et al., 2016; Vial, 2019). Moreover, there is ample evidence that many DT processes have failed due to the lack of leadership skills rather than the primary lack of knowledge about digital technologies (Burke, 2008; Deloitte, 2017).

The general importance of leaders for the success of DT seems to be well substantiated (e.g., Lenkenhoff et al., 2018; Li et al., 2018; Vasconcelos et al., 2016; Vial, 2019; Voigt et al., 2014). However, the question arises as to which specific leadership competencies and skills companies and public administrations need to successfully manage the fundamental transformation in the digital age. A recent literature review by Eberl and Drews (2021) found that the most mentioned attributes of leaders in the digital age are: being visionary, digitally savvy, highly collaborative, adaptable, and motivating. Other

authors propose that they should act as role models by fostering a supportive attitude toward change (Bondarouk and Ruël, 2008; Paulin, 2014; Sow and Aborbie, 2018) and by ensuring that employees have digital know-how (Day et al., 2014). Additionally, and partially overlapping, the relevance of personal traits such as being visionary, creative, selfless, and flexible has been highlighted (Beeks, 2012; Kane et al., 2019; Mihardjo et al., 2019). The first impression that these are general, conceptually derived wish lists is reinforced by a closer analysis of the literature reviews on this topic (e.g., Eberl and Drews, 2021; Guzmán et al., 2020; Klein, 2020; Vial, 2019).

What is lacking so far, with rare exceptions, are studies that analyze actual job requirement profiles for leaders of the DT and compare them to these conceptual models. Focusing the analysis on what employers and their HR departments—considering their specific company context—have identified as necessary characteristics of an ideal candidate offers complementary advantages over other empirical methods (Harper, 2012). This is even more true for organizations that are at the beginning of their transformation process and need the right managers to avoid failure in lieu of the far-reaching effects of DT and its strategic relevance and cannot afford a miscast. It can be assumed that HRM managers entrusted with this staffing will define the requirements very carefully and obtain external expertise. This focus also allows for a better separation of positions and competencies that are more concerned with technical aspects of DT (e.g. cloud computing, big data analytics) from those that are supposed to orchestrate the necessary and comprehensive organizational change process. Against this background, studies analyzing the content of job advertisements can enrich conceptual and practical knowledge about leadership competencies needed for successfully mastering DT processes.

While there are a few relevant studies that have examined required skills by focusing on certain profiles such as chief digital officers or chief technical officers (Reck and Fliaster, 2019; Singh and Hess, 2017; Wade and Obwegeser, 2019) or have used rather small (Sousa and Rocha, 2019) or national samples (Porfirio et al., 2021; Zeike et al., 2019), to our best knowledge, there is only one study which has analyzed leadership requirements for DT based on job advertisements (Klus and Müller, 2020). However, as it focuses on just one specific region in Germany, generalizability of results may be limited.

Hence, following calls for providing empirical evidence from a larger, international sample (Guzmán et al., 2020; Sousa and Rocha, 2019), the objective of this study is to explore and to categorize leadership competencies for DT which are considered necessary by practitioners. To draw robust conclusions for research and practice, accordingly, 239 job advertisements have been analyzed by means of a content analysis. Additionally, we determined the extent to which skills and traits of leaders sought by practitioners to manage DT match competencies identified in conceptual studies.

The remainder of the paper is structured as follows. In the first section, we revisit concepts related to leadership competencies to develop a conceptual framework for our empirical approach. In the subsequent section, readers are provided with important information regarding the methodology applied. After the presentation of the results, they will be discussed with a special attention to the comparison between the requirements demanded by practitioners and those developed in conceptual studies.

Concepts related to leadership competencies

Task and job descriptions and related skill requirements (e.g., Hackman and Lawler, 1971; Hackman and Oldham, 1976; Mumford, 1972) as well as leadership skills and personality traits (Mumford et al., 2000; Yukl, 1981) have been studied by management scholars for decades resulting in an abundance of both complementary and competing concepts and findings. Although it is not the purpose of this study to provide a comprehensive literature review in the specific field of leadership skills (Yammarino, 2013; Yukl, 2012, 2013), we conduct a selective overview of leadership skill conceptualizations in search of an appropriate framework for systematically analyzing the empirical data collected. After introducing the concept of so-called 21st century skills, a selective overview of general leadership skill conceptualizations is provided.

Twenty-first century skills

As the name of this skill set suggests, the concept of *21st century skills* attempts to derive important leadership and working skills from trends such as globalization, digitalization and increase of knowledge work. To fulfill increasingly complex tasks, organizations need highly skilled workers, which are expected to efficiently generate new knowledge from an exponentially growing amount of available information and effectively apply it to interconnected problems (Van Laar et al., 2017). The 21st century skill sets have been developed especially for educational purposes as to develop basic skills needed in the future to compete on a global scale and have been concerned with national competitiveness. These skill sets have been suggested by different organizations. Among them, Partnership for 21st Century Skills (2008), which proposed three types of 21st century skills: learning skills (creativity and innovation, critical thinking and problem solving, communication and collaboration), literacy skills (information, media and ICT literacy), and life skills (flexibility and adaptability, initiative and self-direction, social and cross-cultural skills, productivity and accountability, leadership and responsibility). Later, it has been suggested to include digital literacy, citizenship, and productivity as important 21st century skills (Voogt and Roblin, 2012).

Even if the concept of 21st century skills is supposed to prepare the next generation for future work challenges, it does not provide an appropriate framework to guide the search for leadership competencies needed for DT. The digital aspects associated with 21st century skills are not yet sufficiently defined and, in contrast to digital skills, 21st century skills are not necessarily linked to applications of advanced ICT, particularly in the business context (Van Laar et al., 2017).²

Leadership competencies: Skills and traits

According to the seminal work of Boyatzis (1982), management and leadership competencies, including traits, self-concept, social role, knowledge, and skills, provide the foundation for successfully performing tasks or achieving goals that require professional expertise and coordination. Organizations that face the challenge of successfully mastering DT need to either train existing staff or hire qualified managers. The latter succeeds

by focusing on proven or to be developed skills and traits, known to foster change processes, rather than difficult to grasp aspects such as self-concept and social role. This is also explicitly reflected in job descriptions and job advertisements which mainly highlight skills and traits which are required or beneficial for the successful completion of organizational tasks (Todd et al., 1995; Verboncu and Zeininger, 2015). For the purpose of this paper, skills and traits are therefore considered as the main building blocks of leadership competencies for DT.

Differentiation and organizing frameworks regarding leadership skills. The scholarly literature on conceptualizing, analyzing, and discussing leadership competencies, that is, skills and traits, is both broad and highly specialized (see, e.g., Zaccaro et al., 2018 for a detailed review). One of the first conceptualizations of skills needed in organizations is the seminal work “Skills of an Effective Administrator” (Katz, 1955) that identified the necessary skills for successful job performance in managerial tasks. Katz (1955: 33) defines skill as “. . . an ability which can be developed, not necessarily inborn, and which is manifested in performance, not merely in potential.” He concludes that the ideal executive, that is, an executive able to cope efficiently with any problem in any organization, needs to possess three skills, that is, technical, conceptual, and human, which can be trained and developed.

Subsequently, there have been multiple attempts to define and categorize skills needed for effective management and leadership (e.g., Boyatzis, 1982; Mintzberg, 1973; Peterson and An Fleet, 2004). Building on previous models and categories of leadership skills, more recently the leadership skills strataplex proposed by Mumford et al. (2007), offers a widely accepted framework which seems to be suitable for categorizing leadership skills required for planning and implementing DT processes. This is all the more the case as Mumford et al. (2007) explicitly emphasize that the effectiveness of leadership skills depends on the characteristics and demands of the specific context: leadership skills needed and effective in one context (e.g., managing a crisis like the current pandemic) are less suitable or even ineffective in a different situation (e.g., building a long-term growth strategy). They further conceptualize managerial and leadership skill requirements as being layered (strata) and segmented (plex) and categorize them in terms of four general categories: (1) *cognitive*, (2) *interpersonal*, (3) *business*, and (4) *strategic skills*. Table 1 provides an overview of the four categories and the respective skills according to Mumford et al. (2007), which will be applied as the organizing framework during the empirical analyses below.

Differentiation of leadership traits. For decades, leadership research has focused on identifying traits that distinguish leaders from followers. In contrast to skills, which can be learned and developed, traits are described as attributes people have and which cannot be learned—at least not in the short-term—and are difficult to change. Traits are commonly defined as “. . . relatively stable and coherent integrations of personal characteristics that foster a consistent pattern of leadership performance across a variety of group and organizational situations” (Zaccaro et al., 2004: 104). Neither the seminal review of leader trait research by Stogdill (1948) nor subsequent attempts were able to find a universal set of traits that consistently differentiates effective from less effective leaders. It

Table 1. The four leadership skill categories according to Mumford et al. (2007).

Cognitive skills	Interpersonal skills	Business skills	Strategic skills
Communication skills	Social perceptiveness	Operations analysis	Visioning
Listening	Coordination	Mgmt. of personnel resources	Systems perception
Ability to learn and adapt	Negotiation	Mgmt. of financial resources	Systems evaluation
Active learning	Establishing relationships	Mgmt. of material resources	Identification of downstream consequences
Critical thinking	Persuasion skills		Identification of key causes
			Problem identification
			Solution appraisal

has become evident that certain personality traits alone do not guarantee leadership success. Early research in this regard points out that leadership traits are only a necessary but not sufficient precondition for leadership success (Lord et al., 1986). It has been further argued and shown that people who possess certain leadership traits need to make use of them by acting accordingly (e.g., formulating a vision, role modeling, setting goals) to be successful (Kirkpatrick and Locke, 1991).

Irrespective of whether traits need to be combined with skills to be effective for achieving certain objectives, they are an important stream of leadership research and the basis of several leadership theories (Conger and Kanungo, 1987; House, 1977; Sashkin, 1989; Yukl, 2013). Although researchers from different disciplines have studied the impact of leader traits on leadership success and have proposed context-specific trait concepts (Goldberg, 1990; Judge et al., 2002; Kirkpatrick and Locke, 1991), determining unique leader traits seems to be of special interest for practitioners (e.g., Cantero-Gomez, 2019). Additionally, the change management literature which is directly related to DT processes emphasizes the relevance of certain leader traits for the success of change initiatives (e.g., Longshore, 1987). For these reasons it is proposed to include leader traits in the analysis of leadership competencies for DT. However, due to the lack of specific frameworks regarding appropriate or required leader traits for strategic change processes an exploratory approach regarding the search of traits that are important in the context of DT is adopted.

Methodology

The current stage of DT, which poses a great challenge for private and public organizations as well as the dynamic demand for the development of new skills and traits as, for instance, reflected in the 21st century skills concept, calls for an exploratory research approach.

Analyzing relevant job advertisements to identify and classify competencies related to leaders hired for managing digital change in organizations allows, among others, to fill the research gap of comparing leadership competencies developed and proposed by scholars with those competencies sought by practitioners. It is a well-established approach applied to investigate skills and traits related to certain organizational and managerial tasks (e.g., Brooks et al., 2018; Cvetić et al., 2017; Debusse and Lawley, 2009; Den Hartog et al., 2007; Harris et al., 2012; Todd et al., 1995). Countering criticism that organizations may be careless and/or inexperienced regarding formulating and advertising clear job opportunities, Todd et al. (1995: 24) found that organizations are “. . . *careful to describe skills they actually require*.” The analysis of job advertisements as a form of secondary data also provides certain advantages especially compared to survey-based approaches. Job advertisements need to be as precise as possible so that they mitigate adverse selection problems, are comparatively unbiased, and can be analyzed over time using established content analysis methods (Todd et al., 1995). Although one cannot rule out other biases such as imitation behavior of HR managers who copy job advertisements from various sources or pre-determination through incorporating ideas derived from conceptualizations, a sufficiently large sample can nevertheless provide a robust understanding of competencies that are regarded as relevant by practitioners (Brooks et al., 2018).

Data collection and sampling

Prior to collecting data on job postings, personal interviews with experts and management consultants have been conducted, which revealed that job advertisements in the field of DT management are mainly published on relevant international job portals such as Monster and Vault, and/or on the management consultants' own career pages. Since one of the leading top management consulting firms, McKinsey & Company, is particularly active in the field of advising clients in DT, Digital McKinsey has been included as a source for job opportunities. Hence, we searched for, selected, and analyzed relevant job postings advertised on Monster, Vault (collaborating with Indeed), and Digital McKinsey from August to September 2018.³ The search words applied regarding job titles included DT manager, business transformation manager, chief digital officer (CDO), digital strategy consultant, and digital project manager.

In the case of Monster and Vault, we used the default settings provided on the site, applying “digital” as the main search term. To filter out job offerings that focus on DT management rather than, for example, product development or network administration, we combined “digital” with the search terms “strategy,” “transformation,” “consultant,” or “project manager.” The *Python* software was used to automatically download the content of job advertisements which matched these search terms for further analysis.

The initial sample consisted of 492 job advertisements. Based on a manual review of the job titles and descriptions by the corresponding author, several job advertisements which, on closer analysis, had nothing to do with DT management were excluded. Software developer, web publisher and domain expert software developer are examples of jobs that were removed from the list. Next, job advertisements were excluded that (a)

were related to very specific and narrow digital activities, (b) did not imply activities primarily related to DT, and (c) were related to trainee positions, all of which by definition do not contain leadership responsibilities. This concerned, for example, job offers such as digital account representative, digital solutions architect, data science consultant, and digital analytics analyst.

As a result of this in-depth assessment of initially selected job offerings we were left with a final sample of 239 advertisements. Of these 239 advertisements, 24 were from firms located in Europe, 213 in North America, and 2 in Australia. Regarding previous professional experience our sample contains 96 advertisements related to senior positions (explicitly calling for more than 5 years of professional experience) and 44 to junior positions (no experience at all or experience up to 5 years required). The remaining 99 job advertisements asked for experience without any further temporal specification and, therefore, could not be assigned. In terms of recruiting employers, our sample includes only private-sector companies of various sizes from a range of sectors, with consulting firms representing by far the largest share (62%).

Data analysis

To analyze the data, a content analysis searching for leadership competencies in terms of skills and traits required (Elo and Kyngäs, 2008; Park et al., 2009) has been conducted. We inductively identified the requirements emerging from the job advertisements and coded the occurring skills and traits using the MAX QDA software. We applied the Gioia method (Gioia et al., 2013), as it allows a reduction of many codes into fewer categories. By means of a first order analysis, the content of 239 job advertisements was coded descriptively, summarizing the relevant text passages and explicit skills, competencies, and traits into short, descriptive words. Out of 2781 coded segments, we generated 144 first-order concepts which formed the basis for further coding and analysis. Following the suggestion of Gioia et al. (1994), we subsequently applied a second-order analysis to categorize the data at a higher level of theoretical abstraction and to create second-order themes that embrace the first-order concepts (Glaser and Strauss, 1967; Strauss, 1987). For instance, first-order terms such as project management and project budgeting were assigned to the second-order theme *project management*. Similarly, the first-order terms data analytics, working with data, collecting, and analyzing data, synthesizing data, and exploring data sources, have been subsumed under the second-order theme *big data analysis*. After checking for category nesting and overlap, 47 second-order themes emerged.

Most of the job advertisements are structured similarly and comprise qualifications, experience, knowledge, and other job requirements. Consistent with the conceptual approach, we eliminated knowledge, because knowledge overlaps with other job requirements mentioned elsewhere in the job advertisements. Like Klus and Müller (2020), we used four dimensions—*qualification*, *experience*, *skills*, and *traits*—to assign the 47 second-order themes. To reduce complexity and eliminate outliers we decided to remove specific job requirements mentioned less than 10 times leaving us with a final total of 29 second-order themes.

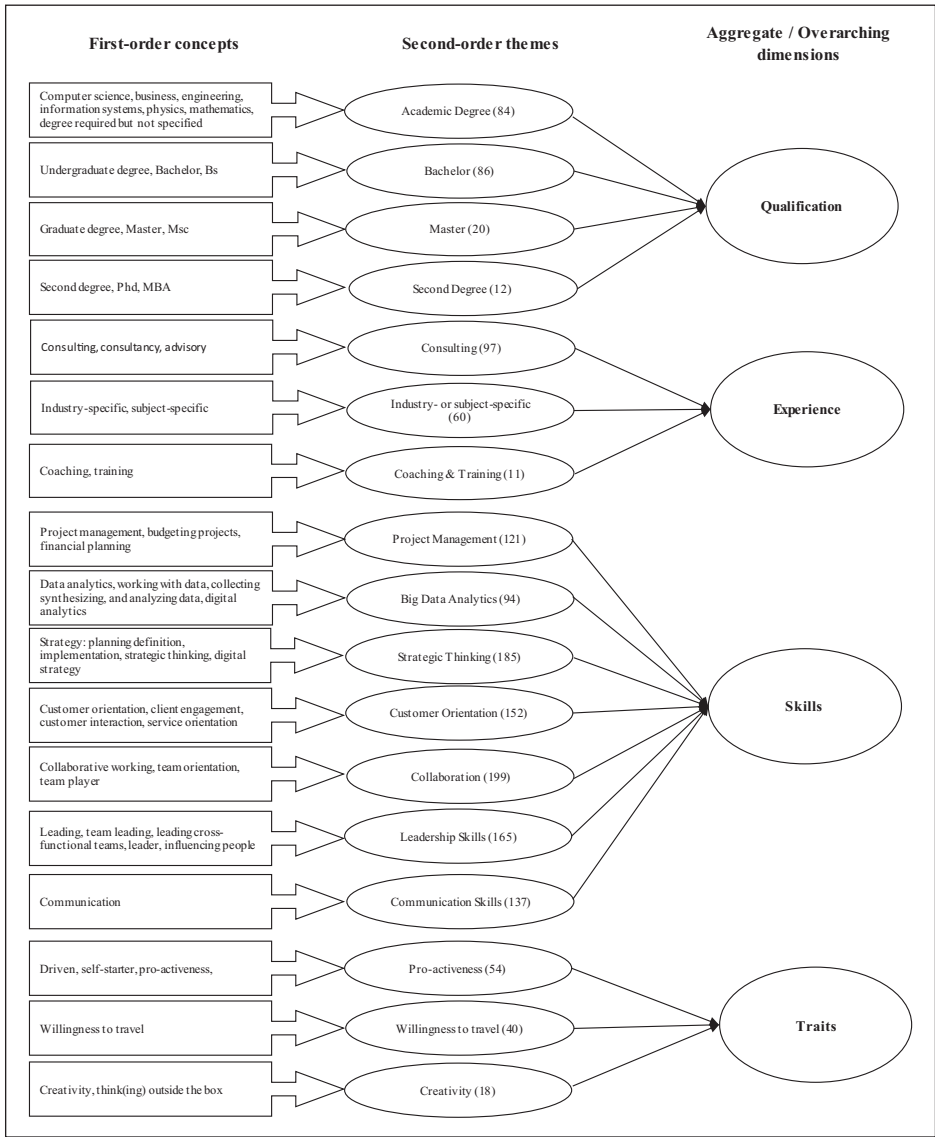


Figure 1. Data structuring illustrated by a selection of second-order themes. Numbers in parentheses refer to the number of job advertisements in which the specific skill requirement was mentioned.

Results

The organization of the data from first-order themes to aggregated skill and trait dimensions is illustrated in Figure 1—using a selection of second-order themes.⁴

Taxonomy development

To develop a taxonomy of leadership skills and traits needed for managing DT successfully, we applied an abductive approach, as suggested by Kalargyrou et al. (2012) and further analyzed the two inductively identified dimensions, that is, skills and traits.

Accordingly, each job requirement related to leadership skills which derived from the content analysis was subsequently assigned to one of the four leadership skill categories *cognitive skills*, *interpersonal skills*, *business skills*, and *strategic skills* (Mumford et al., 2007). First, we assigned communication skills, problem solving, analytical, and presentation skills, as well as multi-tasking, time management, adaptability, the ability to explain complex issues in a simple manner, and listening, to the category of *cognitive skills*. Second, collaboration, leadership skills, and relationship management were assigned to *interpersonal skills*. Third, project management, technological expertise, big data analysis, coding, and programing, agile and scrum, diverse technological skills such as MS Office and Cloud, business-savviness, e-commerce, and design thinking were assigned to the category of *business skills*. Finally, *strategic skills* embrace strategic thinking, customer orientation, change management, business development, and innovation attitude. Table 2 provides an overview of the taxonomy development for the selected leadership skills derived from the job postings. The number in parentheses in the column of second-order themes indicates the frequency this leadership skill was mentioned in the job requirements in the sample.⁵

As previously explained, leadership traits mentioned in job postings for DT leaders are more difficult to cluster and assign due to the lack of a suitable organizing framework. Thus, only a categorization of first-order concepts derived from the content analysis to second-order themes was conducted, which were then subsumed under the single dimension “trait” (Kalargyrou et al., 2012). The four second-order themes proposed are pro-activeness, willingness to travel, creativity, and passion (Table 3).

Descriptive results

Not surprisingly, the analyzed job postings contained a much greater variety of leadership skills than leadership traits—26 second-order *skill* themes compared to four second-order *trait* themes. However, we believe it is surprising that personality traits are mentioned at all in job advertisements relating to DT. It is further noteworthy that pro-activeness is mentioned relatively frequently, which indicates that the change process related to the DT of companies differs from the more reactive process of implementing new legal regulations, for example.

Our study reveals that organizations seeking managers for their DT efforts consider general managerial and leadership skills more important than specific technological skills directly related to digitalization. However, the field of digitalization contains various technologies such as big data analytics, web development, and programing, which is reflected by the high number of mentions (more than 280) in job postings regarding DT. On the other hand, it is notable that key words and terms which are commonly closely related to DT such as Industry 4.0, Internet of things, artificial intelligence, blockchain, or augmented reality did not show up at all.

Table 2. Overview of leadership skill requirements categorized according to Mumford et al. (2007).

First-order concepts	Second-order themes	Overarching dimensions
Communication skills	Communication skills (137)	Cognitive skills
Problem solving	Problem solving (85)	
Analytical skills	Analytical skills (52)	
Presentation skills	Presentation skills (48)	
Multi tasking	Multi-tasking (39)	
Time management	Time management (37)	
Flexibility/Adaptability	Adaptability (13)	
Explain complex issues simply/in a simple manner	Explain complex issues in a simple manner (11)	
Active listening, listening	Listening (10)	
Collaborative working	Collaboration (199)	Interpersonal skills
Team orientation/team player		
Leadership	Team leadership (165)	
Team leading (e.g. leading cross-functional teams)		
Influencing people		
External collaboration/ Stakeholder management	Relationship management (101)	
Project management	Project management (121)	Business skills
Budgeting projects/Financial planning		
Technological expertise	Technological expertise (103)	
Understanding of emerging technologies		
Passion for technology		
Application of technology		
Data analytics / Digital analytics	Big Data analysis (117)	
Working with (digital) data		
To collect and analyze data		
Synthesizing data		
Exploring data sources		
Web development (CSS, Java, HTML)	Coding & Programing (61)	
DevOps		
Programing		
Digital architecture		
Scrum (Product owner)	Agile & Scrum (56)	
Agile methodologies		
Agile environment		
Agile development		

(Continued)

Table 2. (Continued)

First-order concepts	Second-order themes	Overarching dimensions
MS Office Cloud CRM/CMS	Specific technological skills (88)	
Understanding the business	Business-savvyness (27)	
E-Commerce platforms	E-Commerce (14)	
Design thinking	Design thinking (11)	
Strategy Strategy: planning, definition, implementation Visioning Digital strategy	Strategic thinking (185)	Strategic skills
Customer orientation/client engagement Customer interaction	Customer orientation (152)	
Service orientation Change management (transformational change) Digital transformation	Change management (95)	
Business development Innovation, innovative thinking	Business development (45) Innovation attitude (22)	

Numbers in parentheses refer to the number of job advertisements in which the specific skill requirement was mentioned.

Table 3. Overview of leadership trait requirements.

First-order concepts	Second-order themes	Overarching dimension
Driven Self-starter Pro-active Experimental	Pro-activeness (54)	
Willingness to travel Creative Think outside the box	Willingness to travel (40) Creativity (18)	Trait
Passion Interest Enthusiasm Motivation	Passion (14)	

Numbers in parentheses refer to the number of job advertisements in which the specific trait requirement was mentioned.

Table 4. Top five leadership competencies needed for digital transformation.

No.	Leadership competencies	Frequency
1	Collaboration	(199)
2	Strategic thinking	(185)
3	Team leadership skills	(165)
4	Customer orientation	(152)
5	Communication skills	(137)

At the next level of analysis, the most important leadership competencies for DT processes were taken in consideration, revealing additional results of interest (Table 4).

Although one might expect that visionary leadership, foresight, and inspiration would be mentioned most, the five most frequently requested competencies for managing DT within organizations based on our second-order clustering are: collaboration, strategic thinking, team leadership, customer orientation, and communication skills. It should be noted, however, that aligning job requirements mentioned in the job offers to second-order themes leaves room for interpretation due to the lack of clear assignment criteria. For example, if the ability for efficient *time management* ($n=37$) is a separate second-order theme but assigned to project management skills ($n=121$), the latter category would make it to the five most frequently mentioned characteristics.

In the following paragraphs we will highlight several notable insights related to the overview provided by Table 2. Although all skill categories of the Mumford et al. (2007) framework—cognitive skills ($n=432$), interpersonal skills ($n=465$), business skills ($n=598$), and strategic skills ($n=499$)—show a fairly even distribution, *specialist expertise* and *understanding of the business and related digital technologies* appear to be of particular importance for mastering DT.

Regarding the professional requirements subsumed under *business skills*, the aforementioned set of information technology expertise in various fields with a clear focus on applications (e.g., “understanding of emerging technologies”) stands out. The comparatively high number of job postings that explicitly seek candidates with knowledge about scrum and agile organization ($n=56$) may indicate the need for implementing digital solutions that increase an organization’s flexibility. In addition, job postings frequently highlight that successful applicants should have project management experience and skills. This may suggest that a large proportion of the advertised positions are looking for a change agent, rather than the creator of digital business solutions.

However, many companies do not appear to be seeking an operative project manager. Rather they are seeking a manager of DT who can comprehend the strategic dimensions critical to an organization’s successful DT. Many job postings emphasize the need for strategic, long-term, and performance-oriented thinking ($n=185$), making these *strategic skills* the number two skill category. Such a strategic orientation is also supported by the frequent mention of the need for customer orientation in DT ($n=152$). Taking this perspective, DT is not an end in itself but intends to strengthen the strategic positioning vis-à-vis customers. Furthermore, the job advertisements examined explicitly refer to the change management process associated with DT ($n=95$).

This is mirrored in the high number of *interpersonal skills* which are mentioned in the job postings analyzed. DT means a fundamental change in the organization that does not stop at any functional or organizational boundary and, therefore, involves many internal and external stakeholders with whom the manager must work cooperatively. The need to work in a collaborative way across different boundaries and as a respected team player is explicitly mentioned in more than 83% of all analyzed job postings ($n=199$). However, being “only” a team player is not sufficient to successfully managing digital transformation. Additionally, many firms seek candidates that are able to provide direction and motivation to the change teams ($n=165$), making team leadership skills the third most mentioned skill dimension. This is of special interest as most DT teams will be cross-functional.

Finally, a leader of DT initiatives needs distinct *cognitive skills* to be successful. Related to the strong customer orientation and collaborative teamwork, it does not come as a surprise that communication skills are explicitly mentioned as a necessary skill set in more than 57% of all job postings ($n=137$). If requirements such as presentation skills, listening skills, and the ability to explain complex issues in a simple way are grouped together as communication skills, more than four-fifths of all job advertisements contain these skills ($n=206$).

The prevalence of *pro-activeness* ($n=54$) suggests that many companies are not just aiming at responding to new technology developments but are seeking employees who can identify new business opportunities and subsequently implement them, as well as improve internal business operations. Like many empirical studies, it is important to examine both recurring and infrequent mentions to draw meaningful conclusions. Hence, notably, relatively few job descriptions include *creativity* ($n=18$; 7.5%) or *passion* ($n=14$; 5.9%).

Comparison with previous studies

The role of qualifications and prior experience for leadership competencies

Although this study focuses on leadership competencies consisting of a specific skill set and supportive personality traits for managing DT, the job postings examined provided information about qualifications and prior experiences requested for the job (see Figure 1). Our results reveal that while an academic degree, signaling at least a basic knowledge and basic analytical skills, is mentioned as a requirement in almost all cases analyzed (84.5%), a certain level of higher education is often not mandated. From the point of view of managing DT processes, it might have been expected that management-relevant degree programs are preferred, however, results show that no specific discipline dominates DT-related job postings. This suggests that HR practitioners are doubtful that leadership skills on current issues (e.g., digitalization, sustainability, energy transition) are developed in degrees and courses offered by higher education institutions. This finding is in line with findings from related research which highlights similar deficits and resulting challenges of providing employees with skills necessary in the 21st century (Hempsall, 2014; Uhl-Bien et al., 2007).

Highlighting the importance of the specific business context for successful DT initiatives, results show that executives and HR managers explicitly emphasize industry or consulting experience. Two-thirds of all job postings by consulting firms mentioned the need or preference for applicants to have prior consulting experience (97 out of 148), very similar to preference for industry—or subject-specific experience among the other industries (60 out of 91). Interestingly, while employers in this context seem to believe that prior industry experience is beneficial, prior research has shown that it bears hidden costs, too: “. . . in addition to helpful skills and knowledge, workers also carry rigidities that partially offset the benefits of prior related experience.” (Dokko et al., 2009: 63). However, this potential problem applies to most other job advertisements and is not specific to jobs in DT. A possible explanation is, that prior industry experience increases the chance that the selected change manager will focus on developing new business models and improving business processes through new digital technologies, rather than first having to fundamentally understand the business. Moreover, seeking candidates with prior experience may be an indication that many organizations continue to struggle with DT and do not know how to handle this process effectively, ultimately leading to failure (Rogers, 2016; Tollman et al., 2017). To reduce the risk of failure, firms which face the need to further digitalization may prefer to hire external leaders who are already experienced regarding leading DT processes—even if outside of their industry—rather than stick to industry experience.

Comparison with other approaches aiming at capturing leadership competencies

Similar to other management fields, there is an ongoing debate regarding whether academic teaching—and partially research—is lagging behind practical requirements, especially in the field of digital skills (Royle and Laing, 2014) versus whether practitioners do not precisely know what skills and training is required to successfully manage DT processes. Contributing to this debate, we compare our general findings with conceptual studies on leadership skills in the digital age.

Notably, studies originating from the field of information and computer sciences particularly emphasize knowledge in different technology fields of digitalization as leadership skills. Skills related to the context of business applications, change management, and particularly those related to the participation and persuasion of different stakeholders are given lower priority or are not addressed at all. The “*Skills for Digital Transformation*” study is a prototypical example (Hoberg et al., 2015). It surveyed CIOs and other executives responsible for DT worldwide regarding the importance and availability of skills related to certain domains for successfully performing DT using predefined response categories. Actually, response categories were limited to *Entrepreneurship* (importance: 65%—availability: 34%), *Product/Service Integration* (66%–36%), *Business Networks* (71%–42%), *Business Change Management* (84%–46%), and 10 technological domains such as *Blockchain* (36%–24%), *Social Media* (63%–57%), *Internet of Things* (75%–32%), or *Digital Security* (88%–45%). While it is possible that some of these skills were implicitly categorized under *Business Change Management*, several important skills

subsumed under strategic skills or interpersonal skills (e.g., team collaboration and team leadership) in the Mumford et al. (2007) framework are absent or wrongly assumed to be either not critical for DT or anyhow present in organizations.

How do our findings contrast with practice-based conceptualizations and proposals on leadership competencies for DT (e.g., Horlacher and Hess, 2016; Reck and Fliaster, 2019; Rickards et al., 2015; Wade and Obwegeser, 2019), which often focus on chief digital officers (CDOs), chief information officers (CIOs), or chief technical officers (CTOs)? These studies often emphasize general needs such as collaborating with other executives, thinking strategically and being business-centered, pushing customer centrality, designing agile organizations, and having implementation skills. They frequently use aggregated and comparatively vague terminology, which does not appear suitable for establishing a clear requirement profile regarding leadership competencies. However, they provide an idea of how leadership skills and traits need to be directed toward organizational goals and company performance.

Compared to results referred to in literature reviews of digital leadership and leadership competencies for DT (Eberl and Drews, 2021; Guzmán et al., 2020; Klein, 2020; Vial, 2019), our findings provide another interesting insight. It becomes apparent that beyond generic success attributes of leaders (e.g., social intelligence, rewarding, agile, visionary), clearly defining skills and personality traits is necessary to allow operationalization and, consequently, analyses and an assessment regarding job applicants or position holders.

Despite the fact that the term DT is deeply interwoven with technology (Canaday, 2017; O'Neill, 2009; Sebastian et al., 2017) and data (Chen et al., 2017; Dhar and Sundararajan, 2007; Dremel et al., 2017), our results provide empirical evidence that general managerial and leadership skills are at least equally relevant as skills related to specific (digital) technologies. This confirms the assumption that DT needs to be understood as a strategic change process (e.g., Schneider, 2018), requiring leaders to understand the business and their teams, in addition to being digital experts (Furr et al., 2019). Moreover, the importance of traditional leadership and strategy competencies such as team leading, change management, and formulating and implementing a (digital) strategy supports the assumption that change processes that constitute DT are not that different from other organizational change interventions (e.g., reengineering, mergers and acquisitions), allowing to nurture from the established knowledge in the field of strategic change management (Bohnsack et al., 2018).

Finally, how do the results of our study reflect what has been conceptualized as 21st century skills? The concept of 21st century skills initially aims at changing the way students at all ages are prepared in a better, more appropriate way for the future of work increasingly dominated by information and communication technologies, technical devices, and software programs (Dede, 2010; Levy and Murnane, 2004). On the one hand, 21st century skills also include megatrends and grand societal challenges such as globalization, inequality, or climate change, too (e.g., reflected for example in global awareness, civic literacy). On the other hand, 21st century skills do not strive to identify and systematize leadership competencies that ensure successful DT (Van Laar et al., 2017). Hence, while there are certain overlaps in the requirements, for example, regarding collaboration and communication skills, or comprehending information technology

contexts (Dede, 2010), significant gaps, for example, with regard to team leadership, project management or strategic skills are obvious. In other words, not every student will be responsible for managing the DT of organizations. Leadership skills are just one category among many others subsumed under *life skills* according to the Partnership for 21st Century Skills (2008). However, it is known from the leadership literature that leadership success, also in the digital age, depends on the qualifications of those being led (Dvir and Shamir, 2003; Yukl, 2013; Zhu et al., 2009): “*Skills are important for effective performance of leading and following functions.*” (Malakyan, 2014: 12). As such, the 21st century skills initiative aiming at improving the way next generations are prepared for the digital challenges of the future, has an indirect impact on DT. The next generation will have less difficulties to comprehend the opportunities and threats of digitalization and one may expect less explanation effort and less resistance to change. However, such a proposition needs to be further investigated, as the Generation Z is entering the world of work who needs effective onboarding programs, and in view of inter-generational conflicts (e.g. Chillakuri, 2020; Scholz, 2014).

Comparison with another study investigating job advertisements

In comparison with the findings of Klus and Müller (2020), who applied a similar methodological approach, our results show a number of interesting differences and commonalities that might be worth to be further explored by future research initiatives.

To start with, regarding the most important skills and traits between different management levels, Klus and Müller’s (2020) findings do not differ from those shown and described above. Despite some differences in the methodological design, sample and, hence, analyses most of Klus and Müller’s finding seem to also apply to our study. However, the high prevalence of interpersonal skills (e.g., collaboration, team leadership, stakeholder management) and strategic skills (e.g., strategic thinking, customer orientation, change management) which we found in our sample is not reflected in their study. These skills may be hidden in categories, such as *subject specific experience*, *subject specific knowledge*, and *leadership experience*, which are top ranked in their study.

Secondly, we find support for our findings regarding the inclusion of personality traits in job advertisements for DT leaders. Klus and Müller (2020) refer to increasingly differentiated traits such as individual flexibility, commitment, or stress tolerance. In contrast, *pro-activity*, which is frequently addressed in our study, does not appear. We conclude that both studies support the claim that traits continue to be important to employers and may be experiencing a new era with digitalization (Beeks, 2012; Fitzgerald et al., 2013; Hoberg et al., 2015; Kane et al., 2019).

Finally, while both studies proved how important communication and presentation skills are to practitioners and HR managers, for example, they differ in certain respects. In Klus and Müller’s (2020) German sample, for example, intercultural competence and the leader’s self-motivation are attributed a significant role. In our international sample, the ability to lead cross-functional teams is highlighted, and we only found a marginal reference to self-motivation of the leader. This difference could therefore be the result of cultural context or, more broadly, by the institutional context in which the companies operate (Peng et al., 2009).

Discussion and conclusions

The objective of this study is to identify and to categorize leadership competencies which are considered by practitioners as necessary to successfully mastering DT. For this reason, requirements outlined in 239 job advertisements of international firms involved in DT processes have been analyzed.

We found that a great variety of leadership skills and traits are expected to ensure successful DT in addition to technical skills and specific knowledge regarding hard- and software specifications. The strong emphasis on communication skills, cross-functional collaboration in teams, strategic thinking, and customer orientation bears a strong resemblance to other strategic change processes. It seems to reflect the fact that digitalization and an advanced comprehension of digital transformation concepts concerns a process of change that has been ongoing for decades, even if it has recently become more dynamic and extensive. Digital leaders as they are frequently labeled are expected to orchestrate a team of technology experts (e.g., regarding Internet of things, artificial intelligence), business experts and entrepreneurs, and HRM experts (e.g., overcoming resistance, employee participation). In this regard, our research is in line with other authors claiming that people, not technology, drive DT and that organizations require leaders and not necessarily technological specialists for managing the complex changes that comprise an organization's DT (Furr et al., 2019; Vasconcelos et al., 2016; Vial, 2019; Voigt et al., 2014).

Theoretical contributions and practical implications

Considering its conceptualizations and empirical results, our study offers an important contribution for a better and more complete comprehension of leadership competencies, that is, skills and traits, that have an impact on the success of DT processes.

First, by complementing the comparatively underdeveloped empirical coverage of actual job requirements, we contribute to a holistic approach to the evaluation of such competencies. Our findings can be used for comparisons with results derived from different sources, samples and methods applied. Furthermore, they provide a reference point for conceptual studies to compare certain job skills and personality requirements.

Secondly, by applying and transferring the leadership skills strataplex proposed by Mumford et al. (2007) to a contemporary and novel context, we were able to highlight opportunities and limitations of using it as an alternative organizing framework.

Our study also yields important practical implications. First, it offers a systematic overview of skills and traits discussed in the scholarly literature as well as of those sought by organizations facing DT. A systematic overview of the latter skills and traits was not publicly available and can be used to check and complement job descriptions and advertisements currently used. Secondly, it may help to mitigate assumptions that successful DT processes are only possible through hiring technological experts, as it highlights the importance of a collaborative and strategic approach, strong leadership, a high level of customer orientation, and excellent communication skills. Thirdly, our study may be used by practitioners to increase their awareness of the virtue of utilizing conceptual frameworks which may surface skills and traits currently neglected when drafting job descriptions and job calls.

Limitations and future research ideas

Given the complexity and variety of skills and traits listed in job advertisements, as well as the numerous research streams providing in-depth analyses and findings, this study has certain limitations beyond what has been previously discussed.

On the conceptual side, selecting one organizing framework over another will impact the results and their interpretation as the comparison with Klus and Müller (2020) has revealed. As long as there is no generally accepted systematization approach for task-relevant skills and traits specifically for managers of organizational change which accounts for general leadership competencies on the one hand, and for context-specific, partially unknown or neglected skills (keyword: 21st century skills), the use of alternative approaches is necessary. This offers a variety of opportunities for future research in this important field.

A further limitation, although inherent to empirical studies, refers to the selection of the sample and databases used. Excluding other recruiting websites (e.g., The Ladders—for experienced managers or Scouted—for recent college graduates) and channels besides Vault (Indeed) and Monster and other—particularly specialized consulting firms (e.g., Gartner), leading accounting firms (e.g., Deloitte), and software firms specializing in DT (e.g., Oracle/PeopleSoft)—besides McKinsey Digital may impact the generalizability of our results. Since complete coverage cannot be achieved in any case, this limitation results in the call for comparable studies of job vacancies to examine and supplement the existing results. Our study is, so to speak, a first robust push in this direction. The same applies to broadening the geographical (e.g., EU, Asia) and temporal sample as to allow for cross-national, cross-cultural comparisons or to analyze developments and changes over time.

As in other studies, limitations may also apply to the method and analysis. In addition to the advantages mentioned at the beginning, certain disadvantages of an analysis of job advertisements related to its reliability and validity must also be considered (e.g., Harper, 2012). It may be argued that a significant number of jobs might not be advertised externally but are filled internally (Croneis and Henderson, 2002). This is certainly true, but it can be assumed that in this case, too, requirement criteria based on a DT manager's job descriptions are used and that in a well-managed company with functioning HR processes, these will differ only insignificantly from those used for external tenders. Furthermore, as we have already mentioned above job postings depend on the ability of HR managers and recruiters to identify and translate requirements of the actual position into explicit skills and traits (Ahmed, 2005) and may additionally suffer from imitation behavior. The application of complementary empirical methods may be appropriate. For example, future research could analyze how companies, their HR departments, or HR consultancies have derived the required qualifications in the context of DT.

Finally, limitations with respect to the validity and generalizability of our results could result from the coding and categorization of the skills and personality traits. Caused by a lack of objective criteria, this process needs subjective interpretation as demonstrated among others through the comparison with the study of Klus and Müller (2020). This inherent coding and categorization issue applies even more when using content analysis software packages, as the ease to use hides the “carbage in – garbage

out” problem. Particularly, if terminological nuances remain unrecognized and scholars predominantly rely more on quantitative counts. For instance, using too narrow or ambiguous search criteria may unconsciously exclude relevant job advertisements and preselection of search terms by individual researchers may lead to false negatives and false positives. Additionally, general models and organizing frameworks are limited and unable to account for the peculiarities of rather specific and novel tasks and job requirements as in the case of leading and managing DT processes. Because we believe the fundamental problem cannot be solved, a more pronounced focus on this step of the analysis is needed. Collaboration among different researchers and practitioners, including independent categorization that is subsequently reconciled and extensively discussed, could lead to increased validity of the results.

This contribution should be seen as a supplement to other studies and practical experience. With other words, our study does not claim that practitioners are better able to identify and to conceptualize skills and traits necessary for leading a successful DT nor does it reflect reality in an all-encompassing way rather than providing a solid basis for comparisons and future alignments.

Future research may take a closer look at contextual aspects based on extended sample sizes, databases, and methodological approaches. It might be of special interest to analyze whether large companies are seeking different leadership competencies for their DT processes than small and medium-sized enterprises. Another interesting future research stream may compare leadership competencies sought by private versus those of public administration organizations. Furthermore, as we know from international management and especially HRM research, there may be systematic differences in the understanding of leadership and change processes between different countries.

Regarding methods, future research may employ triangulation. As already mentioned, qualitative interviews with HRM and DT experts may be used to identify disregarded or underrated skills and traits which may be taken for granted when searching for a DT leader and, hence, are not explicitly mentioned. Future studies could also methodically identify leadership competencies relevant to success by observing or interviewing important stakeholders of actual DT processes such as project leaders, team members, affected employees, top and middle managers, and employee representatives. Future research in the field of organizational change processes induced by digitalization may analyze even more comprehensively the impact of conceptual factors on the success of change processes as outlined in the respective literature. Such factors include the influence of external pressure to reorganize, the extent of internal power games and conflicts, or barriers such as vested interests of middle managers on the selection of the most suitable DT leader or the set of competencies and traits which match certain situations (e.g., Doppler and Lauterburg, 2019; Miller and Dess, 1996; Nippa, 1997).

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Notes

1. While the scholarly literature like Kotter (1990) or Bennis (2009), for good reasons, calls for distinguishing between management and leadership, this article uses the terms interchangeably. In terms of digital transformation, leaders seem to act as change managers by both driving and implementing the strategic change in an organizational context.
2. Similar skill concepts, which are more related to digital technology but consider the leadership aspect less, are the concepts of Digital Skills (Van Dijk, 2005), Digital Fluency (Briggs and Makice, 2012), and Digital Literacy (Gilster, 1997).
3. Monster (monster.com), Vault (vault.com), Indeed (indeed.com), Digital McKinsey (www.mckinsey.com/business-functions/mckinsey-digital/careers).
4. Table 2 provides a complete list of second-order themes.
5. As already mentioned above, requirements with less than 10 mentions in the 239 cases examined have been removed.

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